

Syamantak Kumar

University of Texas at Austin

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University of Texas at Austin

Master of Science in Computer Science (MSCS)

2021-2023

GPA : 4.0/4.0

Indian Institute of Technology Bombay

Bachelors of Technology in Computer Science with Honours

2016-2020

GPA : 9.35/10.0

Research Interests

Statistical Machine Learning, Optimization, Online Learning, Reinforcement Learning

Publications

1. A comparison of open source libraries ready for 3D reconstruction of wounds [paper]
Syamantak Kumar, Dhruv Jaglan, Nagarajan Ganapathy and Thomas Deserno
Oral Presentation at Proceedings of SPIE Medical Imaging 2019, San Diego, California
2. From Machine Translation to Code-Switching: Generating High-Quality Code-Switched Text [paper]
Ishan Tarunesh, **Syamantak Kumar** and Preethi Jyothi
Oral Presentation at Proceedings of ACL-IJCNLP 2021, Bangkok, Thailand

Research Experience

Online PCA for Markovian Data

Guide: Prof. Purnamitra Sarkar | Research Project

March 2022 - Ongoing

UT Austin

- Working as a Graduate Research Assistant on proving convergence of Oja's algorithm for online PCA with data sampled using Markov Chain Monte-Carlo (MCMC) Sampling

Feature Extraction for Gaussian Process Regression

Guide: Prof. Suyash Awate | Bachelor's Thesis

July 2019 - July 2020

IIT Bombay

Gaussian process regression is a non-parametric Bayesian approach to regression, which provides benefits in generalisation from small datasets and provides uncertainty measurements for predictions. We worked on developing a neural-network based feature-extractor to train in an end-to-end manner and experimented its benefits on different problems in the medical domain.

- Trained end-to-end systems for optimisation of regression parameters using **Deep Kernel Learning**
- Posed cell-counting (counting cells in a slide image) as a GP regression problem and implemented different models for feature extraction, achieving an MAE of 8 on VGG cells dataset
- Attempted the MICCAI BRATS 2020 challenge for **overall survival(OS)** prediction from MRI scans
- Trained **U-Net** models with **knowledge distillation** for segmentation, achieving a dice score of 0.92
- Extracted **radiomic** as well as **CNN-based features** and trained the prediction network with a joint regression plus classification loss to converge to a classification accuracy of 40%

Cross-lingual Embeddings For Code-Switched Text

Guide: Prof. Preethi Jyothi | Research Project

Jan-July 2020

IIT Bombay

- Experimented with MBERT embeddings for hindi-english code-switched text, pre-training on a multilingual corpus and finetuning on smaller code-switched text to achieve improvements in perplexity
- Worked on improving training of RNN-based language models on code-switched text, using mixed-review to address the problem of forgetting during finetuning
- Implemented and developed metrics for evaluation of model-generated code-switched text for perplexity, diversity and text-translation quality

Cell Segmentation and Stain Normalisation

Guide: Prof. Suyash Awate | Research Project

Jan-April 2019

IIT Bombay

- Implemented a **dictionary-learning** based technique for performing segmentation of malaria-infected cell images, using Pytorch alongwith Google Colab

- o Modeled stain normalisation of whole-slide images as a dictionary-learning problem and used a **Huber-prior** to obtain smoother cell images, implemented in MATLAB

3D Reconstruction of Wounds | Assessment and Diagnosis

May-July 2018

Guide: Prof. Thomas Deserno | Summer Internship

TU Braunschweig, Germany

- o Developed an android application to extract **3D point clouds** by scanning objects using **ARCore**, Google's platform for building augmented reality experiences on smartphones
- o Experimented with Poisson, Alpha Complex, Point Cloud, Delaunay Triangulation, Marching Cubes and Convex Hull algorithms for **surface reconstruction** from **sparse point clouds**
- o Compared the results obtained from ARCore with the **Structure from Motion (SfM)** technique to review efficacy of smartphones in generating 3D models for wound assesment

Coursework

Relevant Courses (IIT Bombay) : AI and Machine Learning, Automatic Speech Recognition, Fundamentals of Intelligent Learning Agents, Information Retrieval and Web Mining, Digital Image Processing

Relevant Courses (UT Austin) : Natural Language Processing, Approximation Algorithms, Statistical Methods, Reinforcement Learning, Advanced Machine Learning, Convex Optimization, Theoretical Statistics*, Combinatorics and Graph Theory*

* To be completed by Spring 2023

Teaching Experience

- o **Graduate Teaching Assistant** -
 - CS 361S - Network Security and Privacy, with Professor Seth Nielson at UT Austin for Fall 2021
 - EE 461P - Data Science Principles, with Professor Joydeep Ghosh at UT Austin for Spring 2022
- o **Undergraduate Teaching Assistant** - Among the **20 students** selected across all batches for teaching a class of **47** first-year students for the undergraduate course on Quantum Physics. Coordinated with the Physics Dept. to conduct regular **tutorial sessions & evaluate exam papers**
- o **Teaching Assistant** - Selected as a Teaching Assistant for the Sandbox@Alphabet team's "Linear Algebra for Machine Learning" course, engaging in a chat room of 700+ participants and leading small independent break-out sessions with attendees to discuss problems and exercises.

Professional Experience

Google Maps | Auto-moderation of User Generated Content

August 2020 - August 2021

Software Engineer

Google, Bangalore

- o Designed algorithms to generate signals to detect Hijack & Disguise of businesses on Google Maps
- o Trained lattice regression models for abuse classification based on signals provided by Google Search
- o Performed extensive SQL based analyses to derive insights and statistically verify the algorithms
- o Implemented multiple data processing pipelines in C++ using a MapReduce framework
- o Led discussions with cross-functional teams comprising of 10+ members each, to increase the quality of manual operator labels for abuse classification, with a direct revenue impact

Google Maps | Generative Model for User Contributions

May-July 2019

Software Engineering Intern

Google, Bangalore

- o Built a **generative model** to predict the probability of users making a contribution to Maps
- o Modeled user-contributions as a poisson process and performed estimation with **poisson regression**
- o **Inferred labels** on unlabelled user edits by aggregating future contributions on each location
- o Achieved an **improvement in the coverage of edits** (percentage of edits being labelled) while maintaining similar precision and recall as the original model

Offered a **full-time position** for outstanding contributions during the course of the internship

Notable Projects

Adversarial Examples for Keyword spotting

July 2019 - November 2019

Guide: Prof. Preethi Jyothi | Course Project

IIT Bombay

- o Used **AdvGAN** to generate adversarial examples for keyword spotting by modifying training examples

- Enhanced the performance and robustness of **keyword spotting systems** by training on such examples

Top-k Tournament Ranking From Pairwise Preferences

July 2019 - November 2019

Guide: Prof. Shivaram Kalyanakrishnan | Course Project

IIT Bombay

- Designed and empirically evaluated algorithms for tournament ranking from pairwise preferences
- Explored schemes for fully-sequential sampling by modelling it as a stochastic **multi-armed bandit**
- Theoretically proved correctness under the **Probably Approximately Correct (PAC)** framework

Artistic Style Transfer using Image Processing

July-Nov 2018

Guide : Prof. Suyash Awate | Course Project

IIT Bombay

- Implemented a **transfer technique for re-styling images** based on migrating the style from a given image to the content of another image
- Formulated the problem as an **expectation-maximization** optimisation, with alternating updates
- Used iterated reweighted least squares (IRLS) to approximate the exact solution and improve efficiency
- Achieved results comparable to the style-transfer by CNNs, using only **image processing paradigms**

Toonify - A Cartoonifying Application

July-Nov 2017

Guide : Prof. Kavi Arya | Course Project

IIT Bombay

- Designed an android-based application to perform basic **Image-Processing** operations including cartoonifying an input image, generating a pencil sketch, denoising an image and basic image editing
- Performed **Cel-Shading** for cartoonification using the **Canny-Edge Detection** algorithm for edge detection and **Bilateral Filtering** for smoothing the image with the help of OpenCV4Android
- Implemented an algorithm for generating a pencil sketch of an image using **Color-Dodge Blend**

Chinese Checkers AI

Jan-April 2018

Prof. Amitabha Sanyal | Course Project

IIT Bombay

- Used **minimax strategy** with **alpha-beta pruning** to develop an efficient AI for Chinese Checkers
- Developed a testing interface to conduct **AI vs AI matches** to check the efficacy of our algorithm

Compiler Design

Jan-April 2019

Prof. Supratim Biswas | Course Project

IIT Bombay

- Developed a **lexer** for a subset of C-language using flex (lexical analyser generator)
- Designed a **LALR(1) grammar** for the parser and generated Abstract Syntax Trees using yacc
- Generated three-address intermediate and MIPS assembly codes from **Control Flow Graphs(CFG)**

Awards and Scholastic Achievements

- Secured **All India Rank 30** in **JEE-Main** out of over 1.2 million candidates (2016)
- Achieved **All India Rank 370** in **IIT JEE-Advanced** out of over 140,000 candidates (2016)
- Awarded **AP grade** for excellent performance in **Quantum Physics** and **Physical Chemistry** (2016)
- Received **Gold Medal** for being placed among the **top 38** students in **INChO**, HBCSE (2016)
- Amongst the **top 300** students in **INPhO**, Indian National Physics Olympiad, HBCSE (2015,2016)
- Amongst the **top 300** students in **INAO**, Indian National Astronomy Olympiad, HBCSE (2016)
- Among the **top 47** students to attend the OCSC of International Junior Science Olympiad (2013)
- Awarded the prestigious **KVPY Fellowship** by DST, Govt. of India with **All India Rank 17** (2014)
- Awarded **NTSE Fellowship** by NCERT, Govt. of India with **All India Rank 29** (2012)

Extracurriculars

- Attended **Vijyoshi camp** conducted by IISER, Kolkata which serves as a forum for interactions between bright young students and leading researchers in fields of science and mathematics
- Part of School **Table-Tennis** Team which **won** the **District-Level Tournament** conducted by the District Sports Organisation, Rajasthan
- Stood **first** in the city finals of **Aqua-Regia - The Science Quiz** certified as the largest quiz at one location by the Guinness Book of World Records